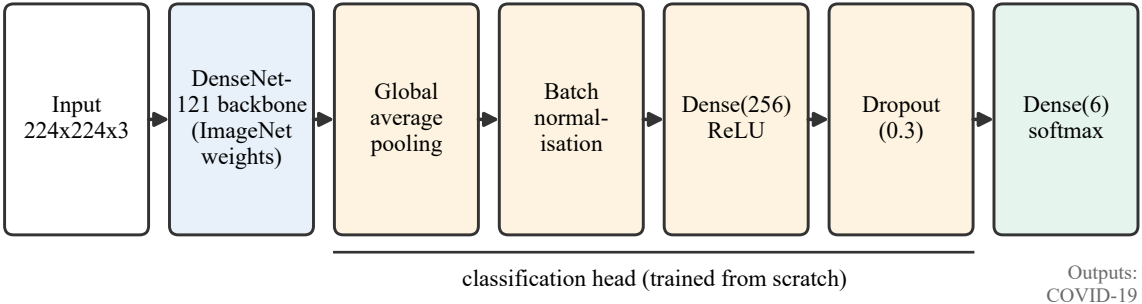
Six-class DenseNet-121 classifier (single-label, mutually exclusive classes)



Outputs:
COVID-19
Normal
Pleural Effusion
Pneumonia
Pulmonary Nodule / Mass
Tuberculosis

Class-weighted cross-entropy, batch size 32, seed 42;
checkpoint = best validation macro-F1

Phase 1: head (2 epochs)
Backbone frozen; Adam, lr = 0.001
Only the head is trained

→

Phase 2: fine-tune (up to 14 epochs)
Last 30 backbone layers unfrozen; Adam, lr = 1e-05
Early-stopping patience 3